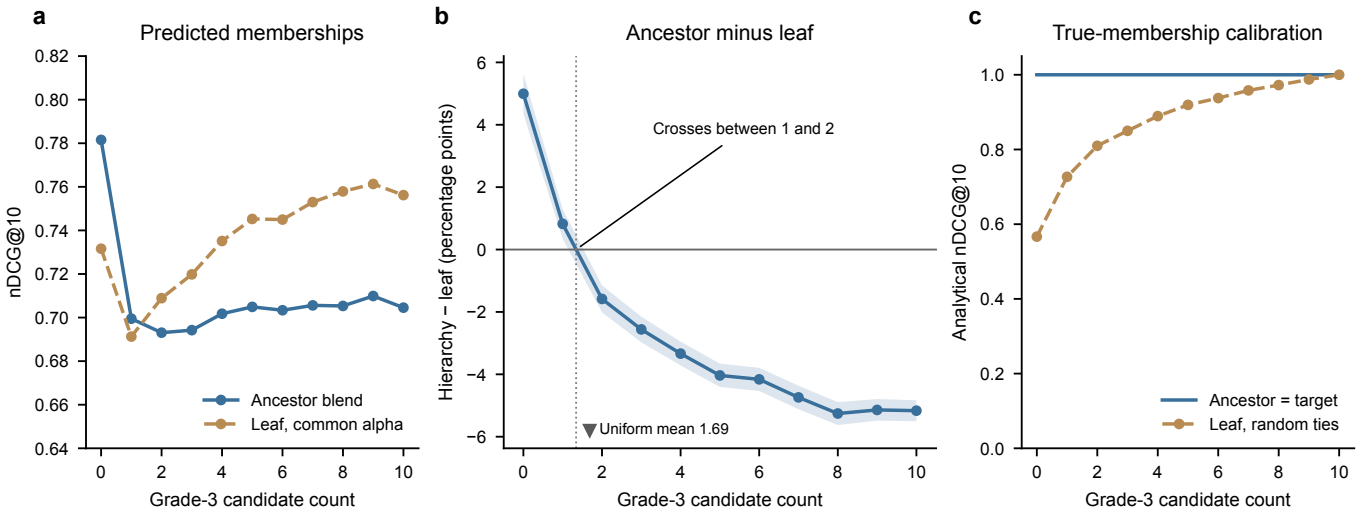
a
Predicted memberships
nDCG@10
0.82
0.80
0.78
0.76
0.74
0.72
0.70
0.68
0.66
0.64
0
2
4
6
8
10
Grade-3 candidate count
Ancestor blend
Leaf, common alpha
b
Ancestor minus leaf
Hierarchy − leaf (percentage points)
6
4
2
0
−2
−4
−6
Crosses between 1 and 2
Uniform mean 1.69
Grade-3 candidate count
c
True-membership calibration
Analytical nDCG@10
1.0
0.8
0.6
0.4
0.2
0.0
Ancestor = target
Leaf, random ties
Grade-3 candidate count